CS7641 - Hypothesis Report Practice

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A. Question 1

As a start, provide background context to help orient the significance. Who are the authors? How might they contribute to the larger field of paper's topic and domain? What is ICML? What is the scope of the conference? How does this conference compare to the larger landscape of machine learning? What is the Test of Time award? You should provide context that gives broader understanding on how the paper might contribute to the field. Please provide 400 words or less.

The authors of the paper "Poisoning Attacks against Support Vector Machines" are Battista Biggio, Blaine Nelson and Pavel Laskov. As most machine learning algorithms assume that their training dataset comes from a natural or well-behaved distribution, the attacker can easily inject malicious input into the model and increase the test error. Their paper aims to demonstrate how poisoning attacks on Support Vector Machines (SVM) can change its model decision function due to malicious input and generate malicious data to the users. Data scientists, software developers and cybersecurity developers can work together to apply security coverage over the machine learning model to minimize the risk of malicious attack and test error.
ICML refers to the International Conference on Machine Learning. The scope of this conference is to gather professionals dedicated to the advancement of the branch of artificial intelligence known as machine learning. This conference presents and awards cutting-edge research submitted by professionals worldwide on all aspects of machine learning. Professionals, graduate students and researchers can learn and understand the impact from reports submitted to ICML for their own reference and apply to their daily work. These reports will be reviewed every ten years for the Test-of-Time award celebrating the lasting impact a paper has had over the past ten years.

B. Question 2

Much like Rule 6 in Ten simple rules for structuring papers by Konrad Kording and Brett Mensh, what is the gap the authors provide? Please provide 250 words or less.

The authors explained in the Introduction that in the world of rapidly growing machine learning techniques, many existing learning algorithms for security-sensitive tasks could still be exploited. As adversaries can be adaptive, they will adapt their approaches and change their behaviors either by evading or misleading machine learning. Even though there are several types of threat for adversarial data manipulation, the authors chose to focus on poisoning attack that directly corrupts training data. In particular, the authors review the family of poisoning attacks against SVM and how an attacker can inject malicious data into SVM reducing its classification accuracy significantly.

C. Question 3

What is the hypothesis of the article? Please provide 100 words or less.

The authors claimed that the SVM can be systematically poisoned where the attacker could inject carefully crafted malicious data into the model training dataset such that the SVM's classification accuracy decreased significantly. The authors hypothesized that even small malicious injection into training data could lead to a significant increase in the classifier's test error. This was because most machine learning algorithms assumed that the training data was natural and independently drawn from a data distribution. However, this may not hold in security-sensitive environments.

D. Question 4

What specifics from the article do you personally need to look up for understanding the claims? How will this information provide context or focus? This can include background, notation, and citations. You must include at least two peer-reviewed citations from an outside source to receive full points. This can be from the citation section or outside source. An LLM search is not a peer-reviewed source. Double check that the article is peer-reviewed. The citation must be in IEEE, APA, or MLA format. Please provide 250 words or less.

There are several specific elements that I personally need to research in order to understand and evaluate the claims stated in the paper. Firstly, understanding how the knowledge and capabilities of an attacker are defined is crucial, because cyber security claims in adversarial machine learning are only valid relative to formal adversary model and threat assumptions. An article by Biggio et al (2013) provides a foundation and peer-reviewed framework, which would help readers contextualize if the paper's assumptions are realistic or overly optimistic. This is because their framework defines adversary goals, knowledge, and capabilities in attacking machine learning. Secondly, it is necessary for the readers to be familiar with the accepted evaluation protocols on cyber security attacks in order to understand how machine learning performance degradation under attack is measured. The article by Akhtar and Milan (2018) can help assess whether the experiments written in the paper are comprehensive or limited, as they provided a survey of adversarial attacks and defenses that situates similar evaluations within broader aspects of machine learning. Lastly, as gradient-based evasion is highly dependent on assumptions about differentiability and smoothness, readers need to evaluate the notations and optimization objectives, especially how gradients and constraints are determined. By comparing these objectives with established formulations in adversarial learning on machine learning, like those empirically evaluated in pattern recognition articles, it would help readers clarify if the mathematical setup done in the experiments aligns with current standards.

E. Question 5

After reviewing and re-reviewing the article, what limitations do you discern? These can be simple or more complex observations. Please provide reasoning in 250 words or less.

After reviewing and re-reviewing the paper, there are some limitations that I discern. Firstly, the authors' analysis focused only on the SVM algorithm. Although they suggested ideas to the readers, they did not provide any formal extension or empirical validation, and the readers may not be able to transfer their findings directly to other modern machine learning models, such as deep neural networks. Secondly, the authors' experiments were conducted on very small and curated datasets, where their experiments were not suitable for the general real world use case - scalability and attack effectiveness in high-dimensional data environments. Lastly, the authors' discussion on defense towards the SVM learning model was very minimal. While the authors convincingly demonstrated vulnerability on the learning algorithms, they did not offer a number of countermeasures - readers were left with an asymmetry understanding between attack complication and defensive guidance. Readers would therefore have to do further research through other possible papers or depend on their internal research team to deploy a suitable defense mechanism against malicious attack on the learning algorithm.

F. Question 6

What might be next steps to further this research? Think of a couple of different options. There are no correct answers here, however your answers must be grounded in reasoning. Be concise with detail. Limit full response to 250 words.

In order to further this research paper, the authors can extend the malicious attack beyond SVMs. By applying the poisoning framework to other deep learning algorithms, such as ensemble methods or deep neural networks, they would be able to test the generality of the core idea. This could potentially reveal whether margin-based optimization is uniquely vulnerable or poisoning attack is a broader structural issue towards other learning algorithms. Secondly, testing poisoning attack on larger scale, high-dimensional or noisy datasets would assess whether this attack remains effective when computational and statistical constraints dominate. Lastly, authors can also extend to co-design defenses, such as data cleansing techniques or robust loss function, and evaluate how they trade off accuracy with robustness. With the authors presenting the attacker-defender dynamics, it could lead to an equilibrium-style analysis.

G. Question 7

Please then provide a single hypothesis on what you'd like to investigate based on the knowledge gained in this exercise. Please provide 100 words or less.

After gaining the knowledge from this exercise, I would like to investigate on the different defense mechanisms against malicious attack on the learning algorithm. This is because training data are extremely important at the very first step in training the model. Once the machine learning model is corrupted through multiple iterations of malicious injections, users might end up receiving wrong instructions from the model. If the corporate deploys such model on their front-end application for customers to use, it will tarnish the corporate image and a lot of effort is required to build back their reputation.

References

[1] Biggio, B., Corona, I., Maiorca, D., Nelson, B., Šrndić, N., Laskov, P., Giacinto, G., & Roli, F. (2013). Evasion attacks against machine learning at test time. Lecture Notes in Computer Science, 8190, 387–402. https://doi.org/10.1007/978-3-642-40994-3_25

[2] Akhtar, N., & Mian, A. (2018). Threat of adversarial attacks on deep learning in computer vision: A survey. IEEE Access, 6, 14410–14430. https://doi.org/10.1109/ACCESS.2018.2807385